

PROGRAM ASSESSMENT REPORT

Tracking data

CIM code	
Program name	BS-BME
academic year	2024-2025
Program assessment coordinator	Promila Dhar
Assessment plan year	2025
No assessment this year per plan	☐
LO(s) assessed	LO1, LO3, LO4, LO6
Presented at program authority meeting date: <i>Please attach minutes</i>	

Methodology

The following LOs were assessed:

LO1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

The criteria used to **assess LO1** were:

- 1a. States assumptions and identifies unknowns
- 1b. Develops models
- 1c. Solves equations
- 1d. Relates solution to physical meaning

The **artifact** used to assess LO1 was:

- Exam 1 problems are designed to assess the students' ability to solve engineering problems.
The assessed course was BME 453. (see artifacts)

LO3: an ability to communicate effectively with a range of audiences.

The criteria used to **assess LO3** were:

- 3a. Conveys technical information using graphics, tables and charts
- 3b. Communicates technical information in oral format
- 3c. Communicates technical information in written format

The **artifacts** used to assess LO3 were:

- Results and discussion of laboratory reports on absorbance experiments to assess their ability to write technical information, which are aligned with LO 3. The course assessed was BME 315 (see artifacts).

LO4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts

The **criteria** used to **assess LO4** were:

- 4a. Identifies and analyzes ethical concerns within the context of the relevant ethical standards
- 4b. Recognizes the economic, social, environmental, and global impact of new research or technology

The **artifact** used to assess LO4 was:

- A homework assignment on Ethics in Biomedical Engineering. The course assessed was BME 100. (see artifacts)

LO6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

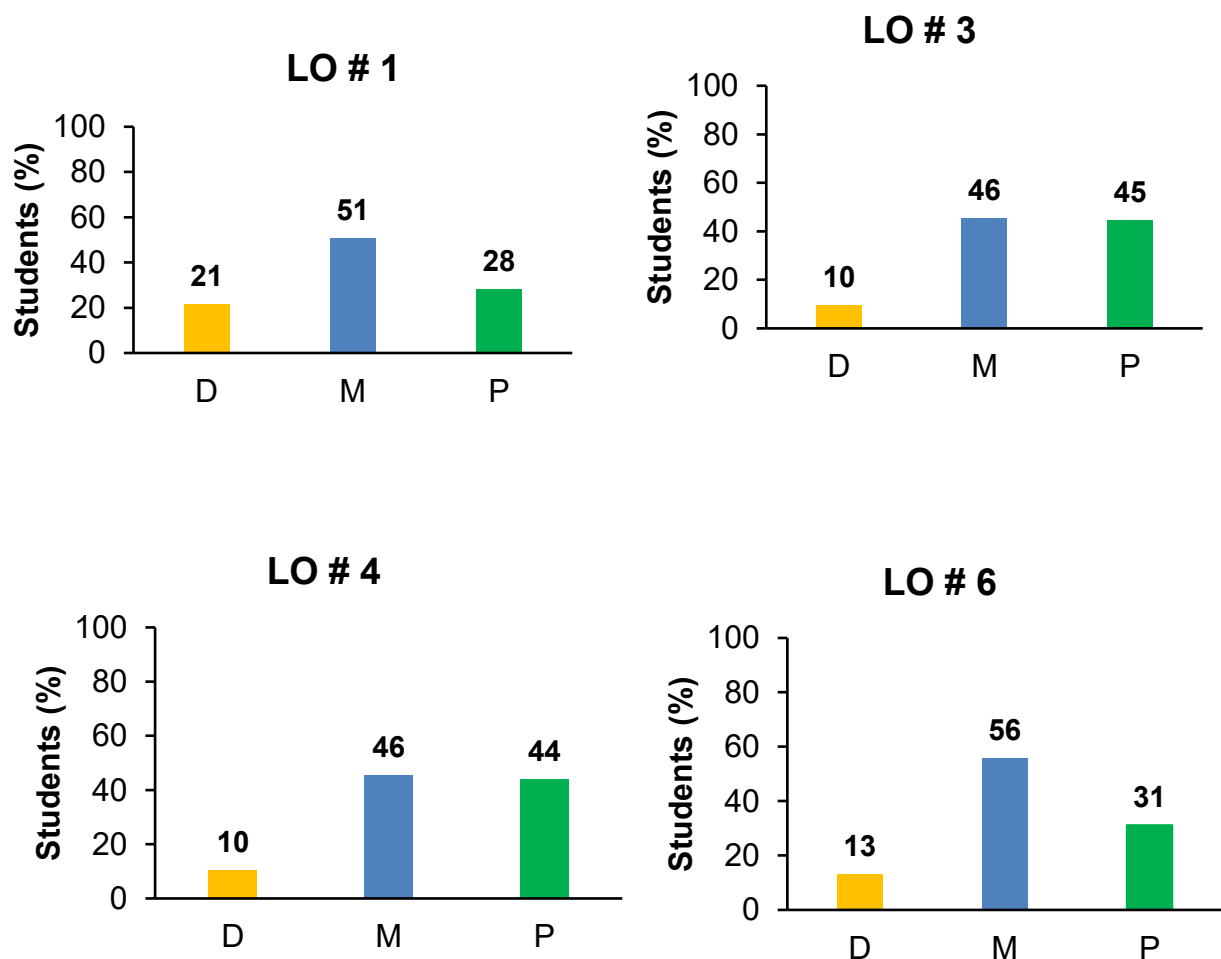
The **criteria** used to **assess LO6** were:

- 6a. Employs a range of manual measurement instruments and computerized data acquisition-based instruments to obtain experimental data
- 6b. Analyzes data sets using basic statistics and presents data using both graphical and textual descriptions
- 6c. Designs an experimental methodology to address a specific objective or test a hypothesis

The **artifact** used to assess LO6 was:

- Assignment to test students' ability to interpret data in Bio-Fluids experiments. The course assessed was BME 320 (see artifacts).

Results



Description of findings

According to the data, the majority of students (>85%) were either proficient or able to meet all LOs 3, 4, and 6. The only assessment that requires further improvement is the one related of LO1, where 21% of the students were deficient.

Continuous Improvement (CI, plans)

The BME department will continue as in previous years, working on the adoption of problem-based assessments in lower-level courses to enhance the students' problem-solving skills, which are assessed in LO1. Another strategic plan is to increase the number of in-class assignments where students can receive feedback on their work in person during the class to enhance students' understanding and mastery of the concepts and skills in all the LOs assessed.

Prior CI plan impact

The implementation of the plan reported in AY 2023 helped maintain the achievement of all the LOs assessed.